

ECG Baseline noise removal techniques using window based FIR filters

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Abstract—This paper deals with the application of the raw ECG signal and best window technique used to design FIR filter by comparing the spectral densities and average power before filtration and after filtration. In this work we have used different ECG databases from Physionet for assessing the quality of the algorithms developed.

Keywords—Average Power, ECG Signal, FIR filter, Spectral density

I. INTRODUCTION

The ECG is a graphical recording of the heart's electrical activity time; therefore it is an important diagnostic tool for assessing heart function. The electrical current due to the depolarization of the Sinus Atria (SA) node stimulates the surrounding myocardium and spreads into the heart tissues. A small proportion of the electric current flow to the body surface. By applying electrodes on the skin at the selected points, the electrical potential is generated by this current and can be recorded as an ECG signal. Fig.1 shows an example of normal ECG trace, which consists of a QRS complex, P wave, and a T wave. The small U wave may also be sometimes observable.

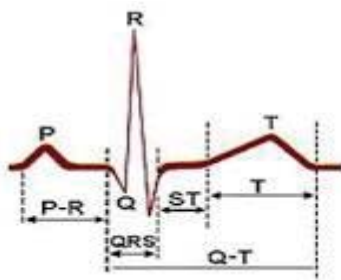


Fig.1.Normal ECG trace.

For diagnostic quality ECG recordings, signal acquisition must be noise free. Since ECG signals are

only of the order of 1 mV in amplitude, the ECG acquisition is susceptible to many types of noises.

The digital filter is used to filter discrete time signals with the ability to modify the frequency response of the filter at any time and it is used in many applications such as data compression, biomedical signal processing, communication receivers, etc [7]. Zhenhe Sun, Wei Yu, designed band-pass filter by using MATAB which possesses superior characteristics of devising filters in a fast and effective way when using window functions originated in it, and the filter was applied to a mixed sine wave signal[9]. Roy, T.K.; Morshed, M presented the design techniques of low pass FIR filters using Tukey window, Kaiser Window, and Gaussian window methods [11].

II. DIGITAL FIR FILTERS

In signal processing, the function of a filter is to take out unwanted parts of a signal, such as random noise, or to specify valuable parts of the signal, such as components lying in a certain important frequency range. A digital filter uses a digital processor to perform numerical calculations on sampled values of the signal for filtering. The processor may be a general purpose computer, or a specialized digital signal processor (DSP) chip. A digital filter is realized by a program stored in the processor's memory. This means the digital filter can simply be changed without affecting the circuitry (hardware). Because of flexibility in their design, digital filters are generally used for filtering. Digital filters are discrete time systems and are characterized by their impulse responses. An impulse response can either have a finite or an infinite duration. The digital filters are used to detect the different signal features on an human heart electrocardiogram signal [10]. Minimum phase FIR filters are known to have certain practical advantages, e.g.

minimum group delay, reduced order for given gain specifications, and lower coefficient sensitivity, when compared with the more common linear phase filters. A finite impulse response 'h (n)' has values extending over a finite time interval from 0 to N and it is zero beyond that interval. The finite impulse response

$$h(n) = (h_0, h_1, \dots, h_N) \quad (1)$$

Has values in the interval (0 to N) and it is referred to as a finite impulse response system or filter of order N. So an Nth order digital filter has an impulse response with a size of (N+1) samples. The samples of the impulse response function h (n) are also called the filter coefficients [2,3].

FIR filters are linear discrete time systems, in which the output sequence is related to the input and the impulse response of the filter by the convolution sum:

$$y(n) = \sum_{m=0}^N x(m) \cdot h(n - m) \quad (2)$$

The summation on the right hand side is a convolution between the input sequence, x (n), and the impulse response of the filter, h (n). The frequency response of an N'th order FIR filter is given by:

$$H(w) = \sum_{n=0}^{N-1} h(n) \cdot e^{-jwn} \quad (3)$$

III. FIR WINDOWING METHOD

FIR filters can be designed using the windowing method. Windows are performed in order to have finite number of samples in time domain for realizable filter design. The ideal filter has infinite number of samples in time domain given in equation (6).

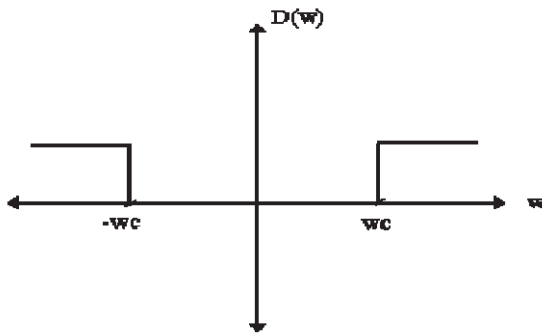


Fig.2.Magnitude response of an ideal filter.

The cut-off frequency is w_c . The characteristic of ideal high pass filter is shown in Figure 2. The

discrete-time impulse response and continuous frequency response are related by the equation 4. The aim is to give the relation between ideal frequency domain filter and its impulse response in time domain and to show the value of windowing method.

$$D(w) = \sum_{k=-\infty}^{\infty} d(k) \cdot e^{-jwk} \quad (4)$$

$$d(k) = \int_{-\pi}^{\pi} \frac{D(w)}{2\pi} e^{jwk} dw \quad (5)$$

The impulse response of filter can be obtained by using the inverse Fourier transform. The filter coefficients will simply be the impulse response samples. The desired response of the low pass filter is given by equation 6:

$$D(w) = 1, \text{ if } |w| < w_c \quad (6)$$

$$= 0, \text{ elsewhere}$$

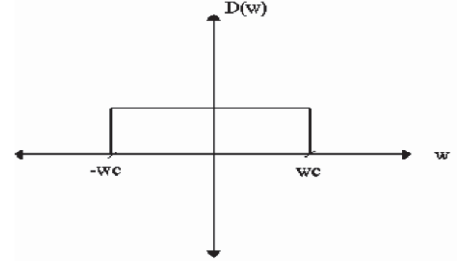


Fig.3.Magnitude response of an ideal window.

A window function from $-w_c$ to w_c is used to show the windowing effect. The impulse response of the ideal window, d (k), is determined in equation 7:

$$d(k) = \int_{-\pi}^{\pi} \frac{D(w)}{2\pi} e^{jwk} dw$$

$$d(k) = \int_{-w_c}^{w_c} \frac{1}{2\pi} e^{jwk} dw \quad (7)$$

$$d(k) = \int_{-w_c}^{w_c} \frac{1}{(2\pi)j \cdot k} (e^{jwk} - e^{-j \cdot w_c \cdot k}) dw$$

$$d(k) = \frac{\sin(w_c \cdot k)}{\pi \cdot k} \quad -\infty < k < \infty$$

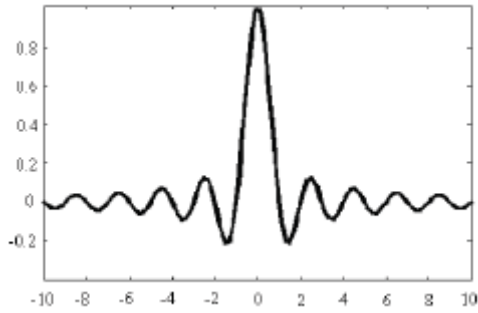


Fig.4.Sinc function d(k) function with k from -10 to 10

In order to compute the impulse response of a finite duration, a window function is employed. The ideal impulse response, $d(k)$, is multiplied by a window function, $w(n)$, which results in a finite duration. There are different windowing functions. The important window functions are rectangular, Hamming, Hanning, and Blackman windows.

A. Rectangular Window

The filter is required to have finite number of values within a definite interval, from $-M$ to M . This is equivalent to multiplying $d(k)$ by a rectangular function given by:

$$w(n) = 1, \quad |n| < M$$

$$w(n) = 0, \quad \text{otherwise} \quad (8)$$

B. Hamming Window

Ringings in the frequency domain is caused by discontinuities in the time function. The rectangular window is replaced by a window function ending efficiently at both ends which will be the reason of the reduction in ripples. The hamming window is an important window function. The hamming window is defined as:

$$w(n) = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right)$$

$$n = 0, 1, 2, 3, \dots, N-1 \quad (9)$$

Where M is the window length and N is the order of the filter. This equation defines the window samples as already shifted (indices from 0 to ' $N-1$ '). So the impulse response of the FIR low pass filter designed using the hamming window is:

$$h(n) = w(n).d(n-M)$$

$$h(n) = (0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right)) \cdot \frac{\sin((n-M)\omega_c)}{(n-M)\pi} \quad (10)$$

The ripples that occur in rectangular windowing in both the pass band and the stop band are virtually

eliminated. Thus, the filtered data will have a wider transition width.

C. Hanning Window

The Hanning window is defined mathematically as:

$$w(n) = 0.5 - 0.5 \cos\left(\frac{2\pi n}{N-1}\right) \quad (11)$$

$$n = 0, 1, \dots, N-1$$

The difference of Hanning window is performed window function. This function is quite similar to the Hamming window.

D. Blackman Window

The Blackman window exhibits a lower maximum stop band ripple in the resulting FIR filter than the Hamming window. It is defined mathematically as:

$$w(n) = 0.42 - 0.5 \cos\left(\frac{2\pi n}{N-1}\right) + 0.08 \cos\left(\frac{4\pi n}{N-1}\right) \quad (12)$$

In the magnitude response the width of the main lobe is wider than that of the Hamming window.

E. Kaiser Window

In this window the side lobe level can be controlled with respect to the main lobe peak by varying a parameter α . Kaiser Window parameter β affects by side lobe attenuation α db [4]. The width of main lobe can be varied by adjusting the length of the filter.

$$B = \begin{cases} 0.1102(\alpha - 8.7), & \alpha > 50 \\ 0.582(\alpha - 21)^{0.4} + 0.07886(\alpha - 21), & 21 \leq \alpha \leq 50 \\ 0, & \alpha < 21 \end{cases}$$

Where $\alpha = -20 \log_{10} \delta$ is the stop band attenuation in db. Increasing β , the amplitude of side lobe gets decreased. Filter order for FIR filter is

$$N = \frac{\alpha - 8}{2.285 \Delta \omega} + 1$$

Here $\Delta \omega$ is the width of the smallest transition region and N is the filter order.

IV. FIR-DESIGN WITH MATLAB

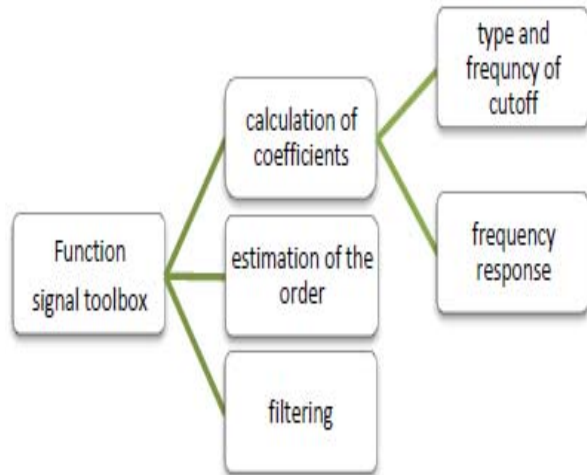


Fig.5.Block diagram of FIR-design

IMPLEMENTATION OF FIR FILTER

In this section, windowing FIR filters with Kaiser, FIR Equiripple filter, Rectangular, Hanning and Blackman functions are performed. As it is discussed basic filter design parameters are stop-band ripple, pass-band ripple and cut-off frequencies in the design of FIR equiripple design, stop-band weight is 0.75 and the desired cut-off frequency is 0.5 Hz and the order of the filter was found to be 320. The transition band of this filter is selected very narrow and approximately 0.1 Hz. It is approximated by taking into account the order of the filter and pass band and stop band ripples. In case of window filters, cut-off frequency at the 3 dB point is 0.5 Hz. The window length in case of Kaiser and rectangular Window is 451 which is selected according to filter order 450(window length is order plus one). But in other windows, order becomes very high and reaches up to 1500 and it increase the phase delay. Figure 6 shows the noisy ECG signal data having Baseline noise.

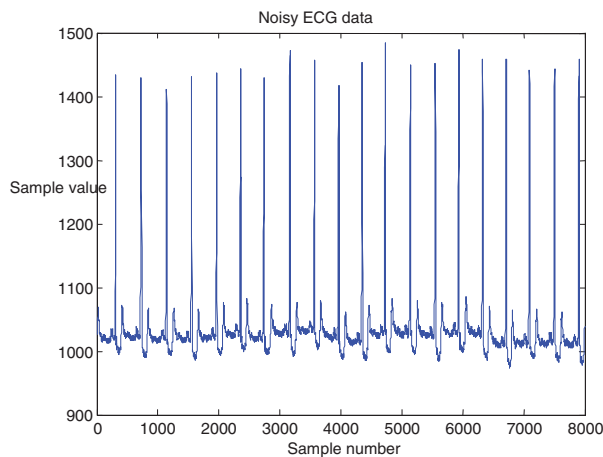


Fig.6.The noisy ECG data having Baseline noise

On applying all above designed FIR filters on ECG sample with Baseline noise as shown in Fig 6, we will get the following waveforms:

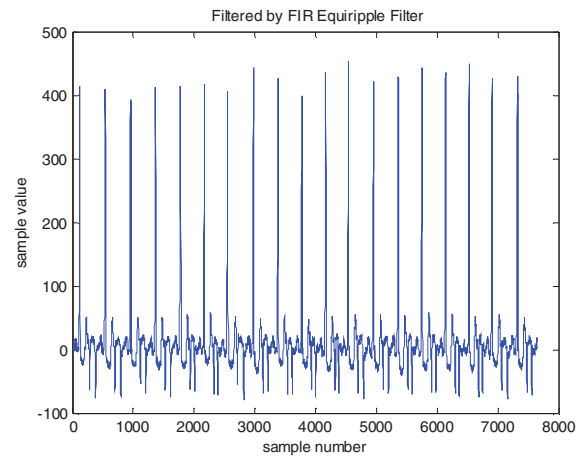


Fig. 7.ECG signal filtered by Equiripple filter

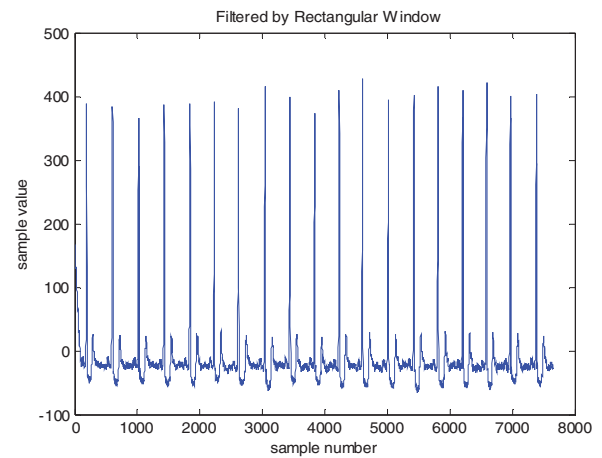


Fig.8.ECG signal filtered by Rectangular window

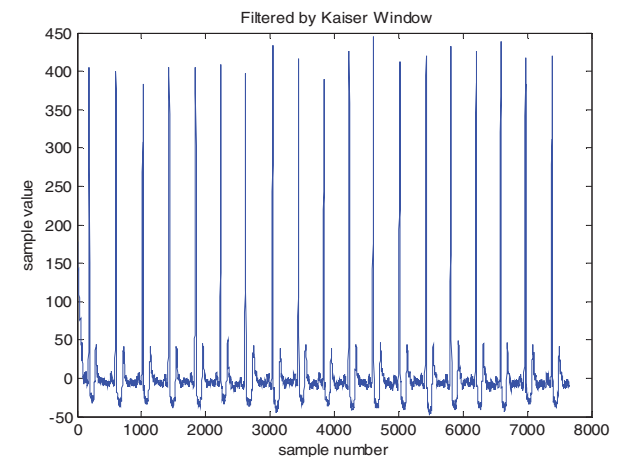


Fig.9.ECG signal filtered by Kaiser Window

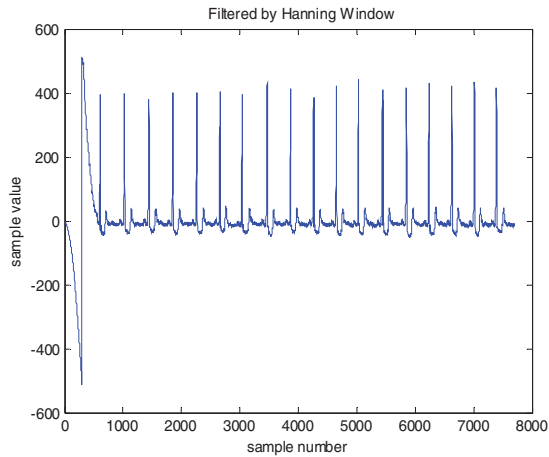


Fig.10.ECG signal filtered by Hanning window

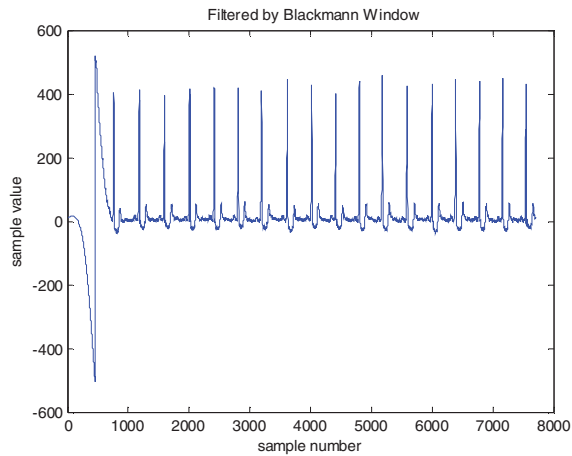


Fig.11.ECG signal filtered by Blackmann window

V. CALCULATION OF PARAMETERS

Spectral density and average power of signal are the two important parameters to check the suppression of Baseline noises. The cut-off frequency used for high pass filters is 0.5Hz. So, the spectral density below 0.5 Hz is calculated before and after filtration of ECG signals. The signal should have good trade-off between average power and spectral density after filtration and also minimum order of filter as possible to avoid the computational complexity.

TABLE I

Filter	Filter Order	Spectral density before filtration (dB/Hz)	Spectral density after filtration (dB/Hz)
FIR Equiripple	320	40.59	29.38

FIR Kaiser	450	40.59	29.57
FIR Rectangular	450	40.59	29.17
FIR Hanning	1200	40.59	30.69
FIR Blackmann	1500	40.59	30.46

Comparison of spectral densities of various filters for Removal of Baseline noise

TABLE II

Filter	Filter Order	Average power before filtration (dB)	Average power after filtration (dB)
FIR Equiripple	320	60.30	30.62
FIR Kaiser	450	60.30	29.74
FIR Rectangular	450	60.30	29.33
FIR Hanning	1200	60.30	30.83
FIR Blackmann	1500	60.30	30.98

Comparison of average power various filters for Removal of Baseline noise

VI. CONCLUSION AND RESULTS

In this paper, noise is reduced from different ECG data signal using FIR digital filter and calculate SNR for various window technique and after comparing these results, spectral densities and average power for Kaiser is better. Table shows the comparison of different FIR filters. The trade-off between spectral density and average power is best among all the FIR filters. But it can also visualize that the waveform got distorted to some extent in case of FIR equiripple filter and the rectangular window. The Kaiser Window and rectangular window is also showing better results at the expense of some more computational load as the order of the filter is large. But in the case of remaining windows i.e. Hanning and Blackman windows, the order of filter easily grow very much high. It increases the number of filter coefficients which increases the large memory requirement and problems in hardware implementation. So, the FIR Kaiser Window filter can be best choice for the removal of Baseline wandering among FIR filters.

VII. REFERENCES

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